

Arun Prajapat

Software Engineer | AI/ML Engineer

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EDUCATION

Indian Institute of Information Technology Vadodara

August 2023 - May 2027

B.Tech in Computer Science and Engineering

CPI: 8.3/10

Vidhya Shri Public School, Didwana (RAJ)

July 2021 - April 2022

Class XII (RBSE) – Science stream

Percentage: 94.8%

EXPERIENCE

AI/ML Research Intern — IIIT Vadodara

June 2026 - Present

- Research and implement advanced deep learning techniques for healthcare applications, focusing on medical image understanding and clinical decision support.
- Conduct model development and experimentation to improve accuracy and reliability on healthcare datasets.

Web Development Intern — Ideal Web Infotech

April 2026 - June 2026

- Developed and deployed Adoreprints, a full-stack personalized gifting e-commerce platform using Next.js, MySQL, Cloudinary, and payment gateway integrations, supporting product customization and order management.
- Developed an AI-powered audio transcription platform using Flask, Faster-Whisper, Resemblyzer, and Groq LLMs for speaker diarization, sentiment analysis, and automated call summarization.

PROJECTS

Multi-scale Pix2Pix GAN-based Diverse Image Colorization  | *Python, Pytorch, GAN, CLIP* April 2026

- Built **CLIP-guided multi-scale GAN** architecture for sketch-to-image colorization.
- Achieved strong visual fidelity with **22.41dB SNR and 0.83 SSIM**, ensuring accurate structural reconstruction.
- Addressed a mean diversity of **0.024** using stochastic latent vectors and AdaIN style injection.

AI Research Assistant  | *Python, LangChain, LangGraph, LangSmith, Streamlit, FAISS*

March 2026

- Developed a multi-agent AI research assistant that generates structured insights from documents using RAG.
- Engineered autonomous workflows for document analysis, search, summarization, and citation generation.
- Optimized retrieval pipelines to improve contextual accuracy and reduce hallucinations in long-form documents.

Dynamic Ensemble for Plant Disease Detection  | *Python, TensorFlow, NumPy*

Jan 2026 - Feb 2026

- Developed a **Mixture-of-Experts** deep learning system to detect plant leaf diseases with **97%+ accuracy**.
- Engineered a custom **Gate Network (CNN)** that dynamically assigned weights to **VGG16, VGG19, InceptionV3**, and **ResNet101** predictions based on input image patterns.
- Integrated **Explainable AI (LIME)** to highlight disease-relevant leaf regions, ensuring model transparency.

Postal Automation  | *Streamlit, Python, OCR, MongoDB*

Oct 2025 - Dec 2025

- Developed an end-to-end postal automation system for extracting destination addresses from postal card images.
- Built a multilingual handwritten **OCR** engine using **R-CNN** and **CTC Loss** for city-name and PIN-code recognition, achieving an average accuracy of 98% across multiple language models.
- Integrated a knowledge graph module using **MongoDB** to improve PIN-code and city-name validation accuracy.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL

Backend & Systems: Node.js, Express.js, Next.js, REST APIs, FastAPI, Prisma, WebSockets, Google OAuth, JWT

Frontend: React.js, Next.js, Redux Toolkit, React Konva, Recharts, HTML5, CSS3, Tailwind CSS

Databases: MongoDB, MySQL

ML & AI: PyTorch, TensorFlow, Scikit-learn, Computer Vision, NLP, Generative & Agentic AI, LLMs, RAG

Tools: Git, GitHub, Postman, Vercel, Railway, Cloudinary, Nodemailer, Razorpay and Cashfree Payment Gateway

Core Concepts: DSA, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, AI, ML

ACHIEVEMENTS & CERTIFICATIONS

GATE 2026 Qualified in Computer Science and Engineering.

Completed Infosys Springboard AI/ML Internship and developed a real-time cryptocurrency forecasting platform.

Earned an NVIDIA Deep Learning Certification covering neural networks and deep learning fundamentals.

Certified in Machine Learning, Deep Learning, and NLP for Developers by Infosys Springboard.

Achieved LeetCode rating of 1681 and CodeChef rating of 1425